

Ved Karamchandani

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EDUCATION

Purdue University, College of Science

West Lafayette, IN

Bachelor of Science in Data Science, Minor in Finance

May 2029

- **Coursework:** Multivariate Calculus, Elementary Linear Algebra, Programming (Java, Python), Discrete Mathematics, Statistics for Data Science, Introduction to Financial Management, Data Science Labs: Calculus

PROFESSIONAL EXPERIENCE

ADL Coordinates

Remote

Avionics Data Research and Analysis Intern

Oct 2024

- Executed a comparative analysis of Aeronautical Information Management digitization across 36 UN ICAO member states, utilizing Excel to systematically rank data on a 3-part maturity scale.
- Tracked the development of the AIM components such as AIP, OLS, SWIM, NOTAM and Charting.

Labdox Private LTD

Pune, Maharashtra

Data Analytics and Web Development Intern

May 2023

- Analyzed and segmented data for over 2,000 undergraduates to identify potential candidates, and designed a dedicated landing page to optimize webinar registration conversion rates.
- Developed and executed a comprehensive marketing strategy for a new AI course and webinar, increasing engagement and brand visibility among target audiences.

CAMPUS INVOLVEMENT

Purdue Solar Racing

Software and Logistics Teams

Sep 2025 - Present

- Currently designing and prototyping the primary driver interface for the solar race car, focusing on real-time visualization of critical metrics (e.g., battery health, motor efficiency, torque and energy consumption) to directly inform in-race strategy.
- Spearheading local sponsorship acquisition by identifying, researching and initiating contact with sustainable energy firms.

LaunchPad Purdue

Mentee

Sep 2025 - Present

- Selected as one of 40 mentees to further develop 'OnTime', utilizing Linear, Lasso/Ridge, and Random Forest regression models on scraped flight data to provide customized efficiency solutions for individual airlines and airports.
- Participated in specialized workshops covering Machine Learning, Web/Mobile Development, and technical case studies, while engaging with industry alumni from Google, IMC, and Roblox during technology conferences.

PROJECTS

HelloWord Hackathon

Sep 2025

- Designed a flight delay predictor dashboard called 'OnTime' that collects data from flights in the United States and predicts the percentage chance of delay using metrics such as weather patterns, maintenance checks and previously delayed flights.

Digital Inclinometer & Height Measurement System | Academic Project

Oct 2025

- Engineered a handheld inclinometer using a Raspberry Pi Pico and MPU-6050 accelerometer to capture real-time acceleration data via I2C communication protocols and developed Object-Oriented MicroPython scripts to implement a moving average algorithm, significantly reducing sensor noise and improving measurement precision.

Multi-threaded Reservation System | Academic Project

Oct 2025 -- Dec 2025

- Collaborated in an Agile team to architect a concurrent client-server reservation system in Java, implementing thread-safe database persistence, Network I/O, and a GUI frontend while ensuring stability via comprehensive JUnit testing.

AWARDS AND TECHNICAL SKILLS

Languages: Java, Python (Pandas, NumPy, MicroPython, Seaborn, SymPy, Matplotlib), R (dplyr, tidyr, lubridate, ggplot2)

Skills: Data Structures & Models, Monte Carlo Simulations, ANOVA Models, Git, Microsoft 365

HSM Business and Entrepreneurship Internship Competition | 1st Place

Dec 2023

University of Waterloo FERMAT Mathematics Contest | Certificate of Distinction

Apr 2024

University of Waterloo HYPATIA Mathematics Contest | Certificate of Distinction

Apr 2024